

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

Year 11 ASSESSMENT TASK 2

JULY 2012

General Instructions

- Working Time - 65 minutes
- Approved calculators may be used.
- All necessary working should be shown for every question.
- Begin each question on a new side of the answer booklet.
- Marks may be not be awarded for careless work or illegible writing.
- For questions 1-5 write answers on the multiple choice answer sheet

NAME _____

TEACHER _____

Total Marks **53**

Section I 5 marks

Section II 48 marks

SECTION I

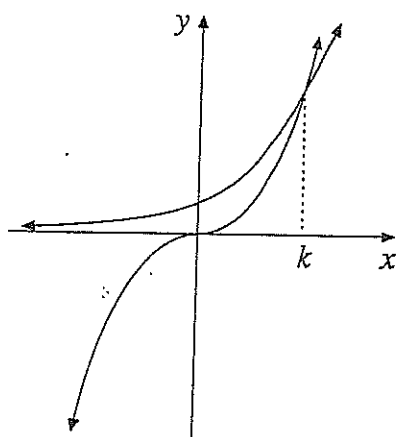
Use the multiple choice answer sheet select the alternative A,B,C or D that best answers the question.

Question 1

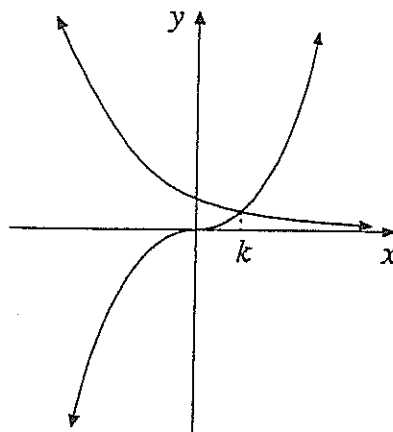
(1 mark)

In which of the following could the value of k be a solution $2^x - x^3 = 0$?

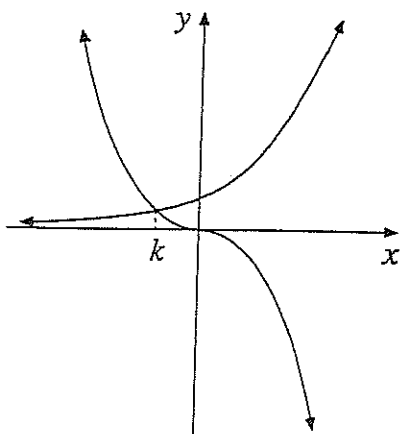
(A)



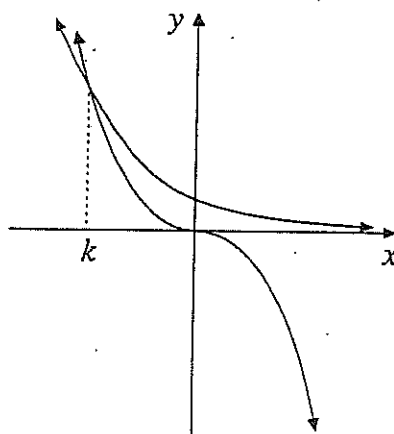
(B)



(C)



(D)



Question 2

(1 mark)

If f is defined by

$$f(x) = \begin{cases} x + 1, & \text{if } x < 3 \\ x + 3, & \text{if } x \geq 3 \end{cases}$$

Find the value of $f(4) - f(2)$

(A) 2

(B) 3

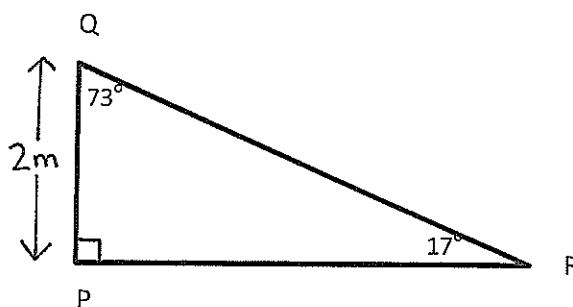
(C) 4

(D) 5

Question 3

(1 mark)

PQR represents a side view of a water skiing ramp. The length RQ in metres is given by



- (A) $\frac{2}{\sin 17^\circ}$ (B) $2 \tan 17^\circ$ (C) $2 \cos 73^\circ$ (D) $\frac{2}{\tan 17^\circ}$

Question 4

(1 mark)

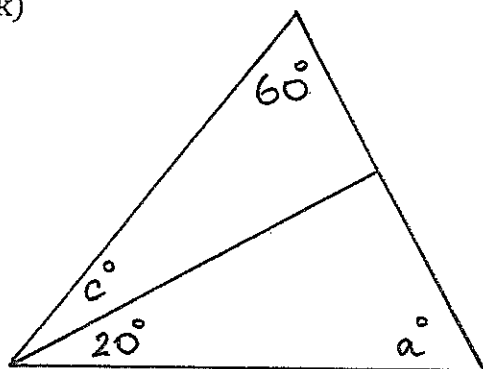
If $\tan y = 3$ and $\cos y$ is negative, then

- (A) $\sin y = \frac{3}{\sqrt{10}}$ (B) $\cos y = -\frac{1}{3}$ (C) $\tan^2 y = \sqrt{3}$
(D) $\sin y = -\frac{3\sqrt{10}}{10}$

Question 5

(1 mark)

not to scale



The value of a in terms of c is

- (A) $\frac{c}{2}$ (B) $c - 20$ (C) $100 - c$ (D) $80 - c$

SECTION II

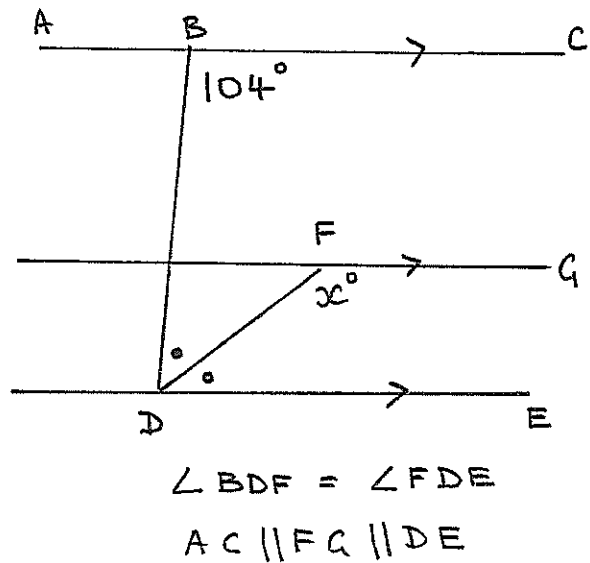
Attempt all questions, starting each question on a new page

Question 6

(6 marks)

- a) Find the value of x
(give full reasons)

not to scale



3

- b) Find the exact values of

i. $\sin(-45^\circ)$

1

ii. $\sec 330^\circ$

1

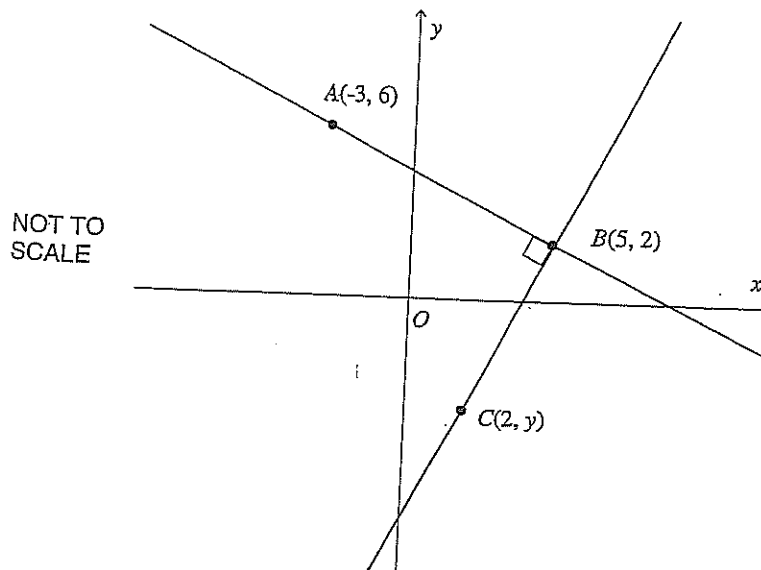
iii. $\tan 420^\circ$

1

Question 7

(12 marks) (Start on a new page)

a)



The diagram shows the origin O and the points $A(-3, 6)$, $B(5, 2)$ and $C(2, y)$.

The lines AB and BC are perpendicular

Copy this diagram onto your writing sheet.

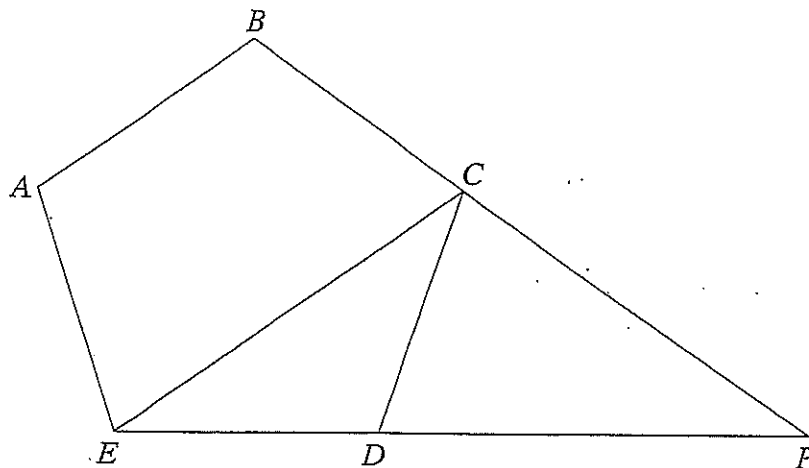
- | | | |
|------|--|---|
| i. | Show that A and B lie on the line $x + 2y = 9$ | 2 |
| ii. | Show that the length of AB is $4\sqrt{5}$ units | 1 |
| iii. | Find the perpendicular distance from O to AB | 2 |
| iv. | Find the area of triangle AOB | 1 |
| v. | The equation of line CB is $2x - y - 8 = 0$
Find the co-ordinates of point C | 1 |
| vi. | The point D is not shown on the diagram. The point D lies on the x axis and $ABCD$ is a rectangle. Find the coordinates of D . | 1 |
| vii. | On your diagram, shade the region satisfying the inequalities $x + 2y \geq 9$ and $x \geq 0$ | 2 |
- b) The point $M(3, 5)$ is the mid-point of the interval AB where A is the point $(-1, 2)$. Find the co-ordinates of B
- 2

Question 8

(6 marks)

(Start on a new page)

- a) The diagram shows a regular pentagon $ABCDE$. Sides ED and BC produced



Copy or trace the diagram into your writing booklet.

- | | | |
|-------|---|---|
| i. | Find the size of $\angle CDE$ | 1 |
| ii. | Hence, show that $\triangle EPC$ is isosceles (reasons required) | 2 |
| b) i. | Sketch $y = x + 2 $ and $y = \frac{1}{3}x + 2$ on the same axes. Indicate where they cross the x and y axes. | 2 |
| ii. | Hence solve $ x + 2 > \frac{1}{3}x + 2$ | 1 |

Question 9

(6 marks)

(Start on a new page)

- | | | |
|----|---|---|
| a) | Solve $\sin \theta = \frac{-1}{2}$ if $0 \leq \theta \leq 360^\circ$ | 2 |
| b) | Simplify $(\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta + \cot \theta)$ | 2 |
| c) | Find the length of the diagonal of a rectangle, if it makes an angle of $63^\circ 30'$ with the shorter side, which is 4.5cm in length.
(answer correct 1 dec. p1) | 2 |

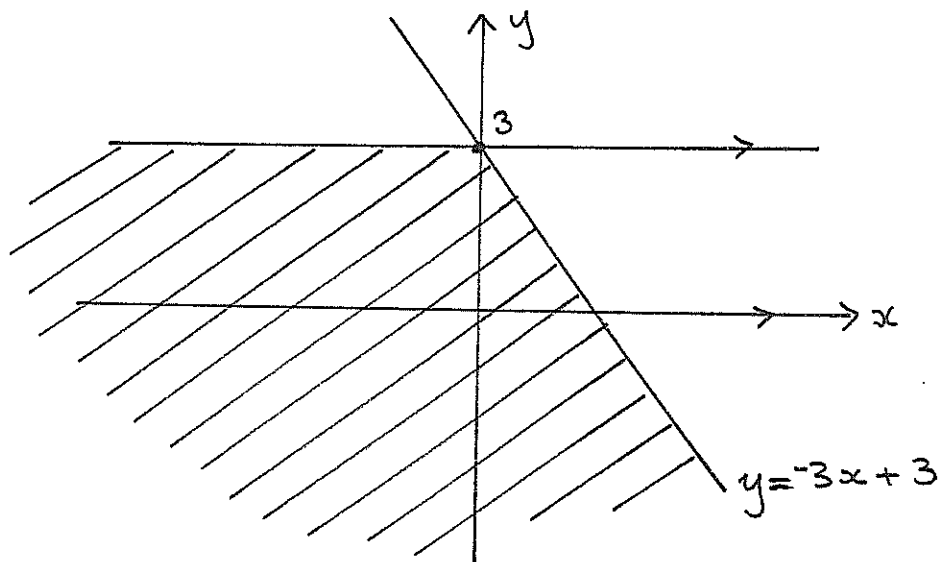
Question 10

(6 marks)

(Start on a new page)

- a) Write a pair of inequalities that describe the shaded region

2



- b) i Find the vertical asymptote for the function $y = \frac{4}{x-2}$

1

- ii Sketch the function (use a ruler). Show all important features
Find and label any points where the curve cuts the x or y axes.

2

- iii State the range of this function

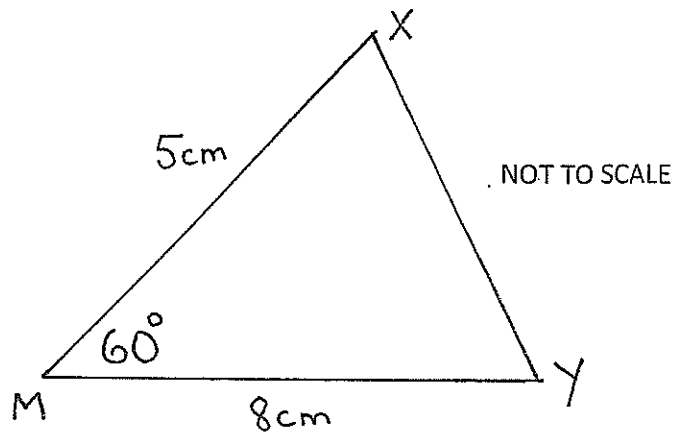
1

Question 11

(6 marks)

(Start on a new page)

- a) Find the exact area of the triangle MXY



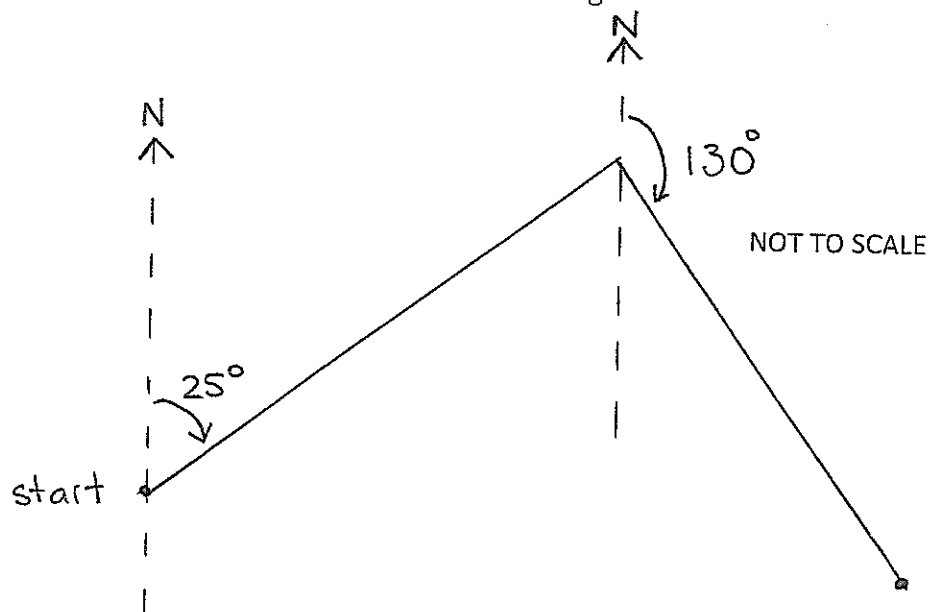
2

- b) A helicopter flies 45km on a bearing of 025° .

It then flies 56km on a bearing of 130° .

- i Show its distance from the starting point is 62km (to nearest km) 2
- ii What bearing needs to be taken to return to its starting point? 2

Give your answer correct to the nearest degree.



Question 12

(6 marks)

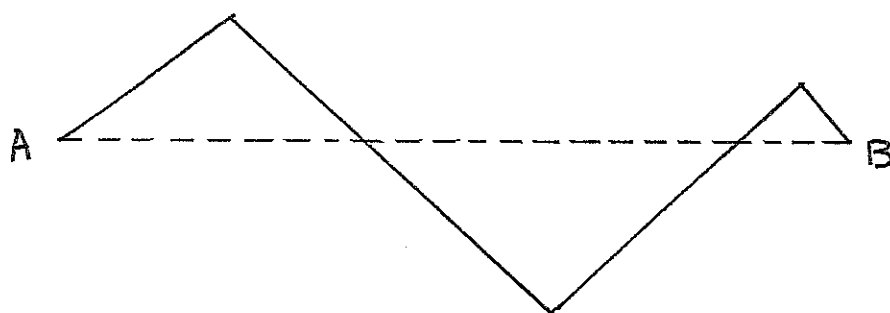
(Start on a new page)

a) Prove $\frac{1}{\sin \theta \cdot \cos \theta} - \tan \theta = \cot \theta$

3

- b) A sailing vessel sets a course alternating between N60°E and S60°~~W~~ E. What is the distance between two points A and B, if the ship sails 4000m.
(answer to nearest metre)

3



NOT TO SCALE

Question 1 A

Question 2 C

Question 3 A

Question 4 D

Question 5 C

Question 6

i) $\hat{D}\hat{E} = 76^\circ$ (co-interior angles $\hat{A}\hat{C} \parallel \hat{D}\hat{E}$)
 $\hat{F}\hat{D}\hat{E} = 38^\circ$ ($\hat{B}\hat{D}\hat{E}$ bisected)

$\therefore \hat{x} = 142^\circ$ (co-interior angles)
 $\hat{E}\hat{A} \parallel \hat{D}\hat{E}$

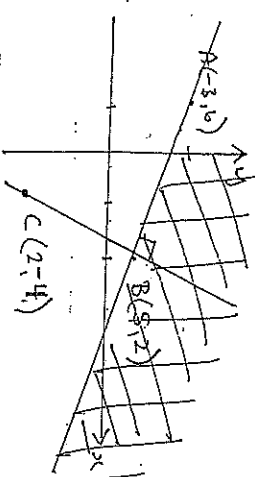
ii) $\sin(-45^\circ) = \sin(360-45^\circ)$
 $= -\sin 45^\circ$
 $= -\frac{1}{\sqrt{2}}$

iii) $\sec 330^\circ = \sec(360-30^\circ)$
 $= \sec 30^\circ$
 $= \frac{2}{\sqrt{3}}$

$= \frac{\sqrt{3}}{3}$

iv) $\tan 420^\circ = \tan 60^\circ$

Question 7



a) $A(-3, 6)$ sub into $x+2y=9$

$-3+12=9$ true

b) $B(5, 2)$ sub into $x+2y=9$

$5+4=9$ true

$\therefore A$ and B lie on $x+2y=9$

ii) $AB = \sqrt{(6-2)^2 + (-3-5)^2}$

$= \sqrt{16+64}$

$= \sqrt{80}$

$= 4\sqrt{5}$ units

iii)

$$P = \frac{|1x_0 + 2x_0 - 9|}{\sqrt{1+4}}$$

$$= \frac{|3x_0 - 9|}{\sqrt{5}}$$

$$= \frac{|x_0 + 2y_0 - 9|}{\sqrt{5}}$$

$$a=1, b=2, c=-9$$

$P = \frac{9}{\sqrt{5}}$ units

iv) $\Delta AOB = \frac{1}{2} \times \frac{4\sqrt{5} \times 9}{\sqrt{5}}$

$= 18$ units

v) sub $x=2$

$y = 2x - 8$

$y = 4 - 8$

$y = -4$

$\therefore C(2, -4)$

vi) $D(-6, 0)$

vii) See diagram

b) let $B(x, y)$

$\frac{x-1}{2} = 3$

$x-1=6$

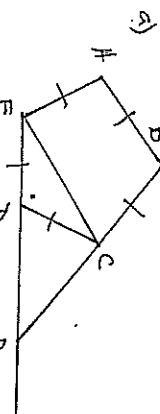
$x=7$

$y+2=5$

$y+2=10$

$y=8 \therefore B(7, 8)$

Question 8



i) $\hat{C}\hat{D}\hat{P} = \frac{360}{5} = 72^\circ$

$\therefore \hat{C}\hat{D}\hat{E} = 108^\circ$ (angles on st. line)

ii) $\hat{C}\hat{D}\hat{P} = \hat{B}\hat{C}\hat{P} = 72^\circ$ (ext angle property)

$\therefore \hat{C}\hat{P}\hat{D} = 36^\circ$ (angle sum triangle)

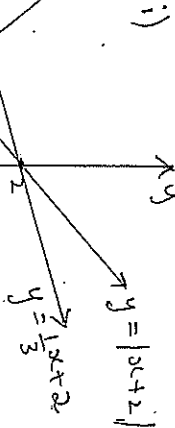
$\hat{C}\hat{E}\hat{D} = 36^\circ$ (ext angle of triangle is equal to sum of interior opposite angles, ΔEBC is isosceles)

ΔEPC is isosceles (2 equal angles)

$\therefore \hat{C}\hat{E}\hat{D} = \hat{C}\hat{P}\hat{D} = 36^\circ$

b)

i)



ii) Find pt intersection

$x+2 = \frac{1}{3}x+2$

$0 = -\frac{2}{3}x$

$x=0$

$\therefore$ Solution $x > 0, x < -3$

$x = -3$

Question 9

a) $\sin \theta = \frac{-1}{2}$

acute $\theta = 30^\circ$

$\therefore \theta = 210^\circ, 330^\circ$

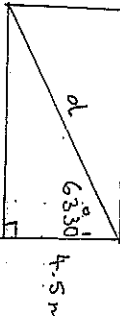
b) $(\cos \theta - \cot \theta)(\cos \theta + \cot \theta)$

$= \cos^2 \theta - \cot^2 \theta$

$= (1 + \cot^2 \theta) - \cot^2 \theta$

$= 1$

c)



$\cos 63^\circ 30' = \frac{d}{6.330}$

$d = \frac{4.5}{\cos 63^\circ 30'}$

$d = 10.1$ cm

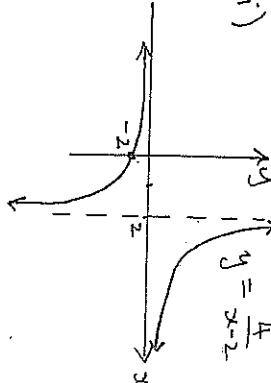
Question 10

a) $y < -3x + 3$

$y \leq 3$

b) i) $x=2$

ii)



iii)

R: all real $y, y \neq 0$

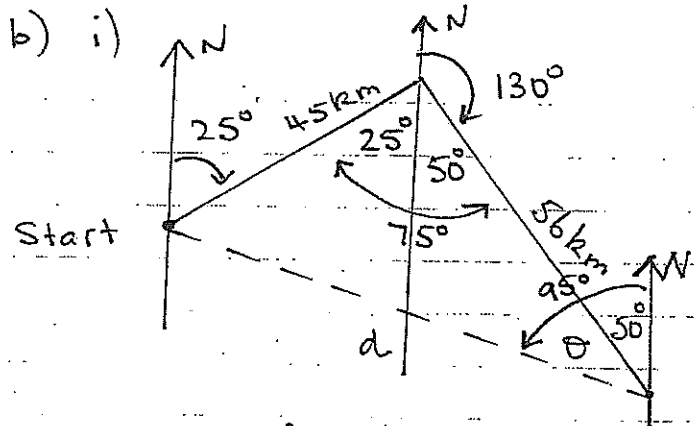
b)

Question 11

$$a) A = \frac{1}{2} \times 5 \times 8 \times \sin 60^\circ$$

$$= 20. \frac{\sqrt{3}}{2}$$

$$A = 10\sqrt{3} \text{ unit}^2$$



$$d^2 = 45^2 + 56^2 - 2 \times 45 \times 56 \times \cos 75^\circ$$

$$d = 62 \text{ km (to nearest km)}$$

ii)

$$\frac{\sin \theta}{45} = \frac{\sin 75^\circ}{62}$$

$$\theta = 45^\circ$$

Bearing

$$\text{OR } 265^\circ$$

$$\underline{\underline{S 85^\circ W}}$$

Question 12

$$\text{LHS} = \frac{1}{\sin \theta \cdot \cos \theta} - \tan \theta$$

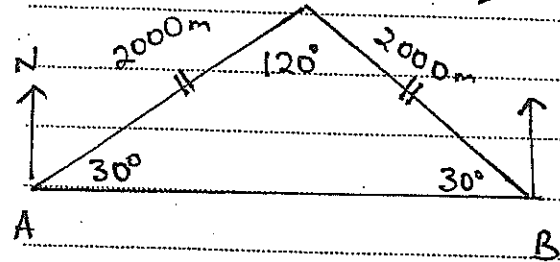
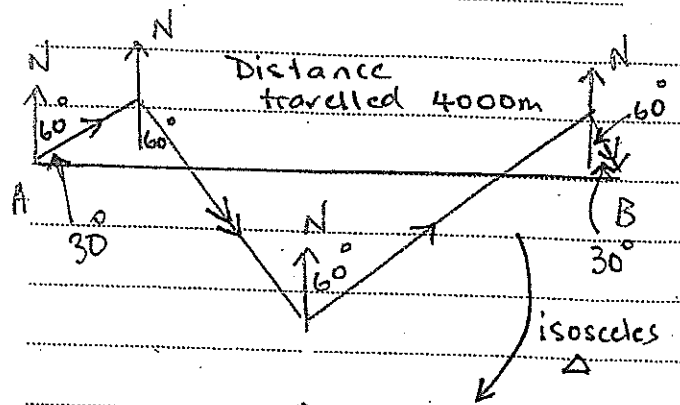
$$= \frac{1}{\sin \theta \cdot \cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta \cdot \cos \theta}$$

$$= \frac{\cos^2 \theta}{\sin \theta \cdot \cos \theta} = \cot \theta$$

$$= \frac{\cos \theta}{\sin \theta}$$

$$= \text{RHS}$$



$$\therefore AB^2 = 2000^2 + 2000^2 - 2 \times 2000 \times 2000 \times \cos 120^\circ$$

$$\therefore AB = 3464 \text{ m (nearest m)}$$